One-Year Kidney Graft Survival Prediction Using Federated Learning and the OPTN Dataset

***Abstract*—Predicting kidney graft survival is crucial for enhancing post-transplant results and assisting with clinical decision-making. Improved post-transplant care and better clinical decision-making can be supported by early one-year graft survival prediction. However, the majority of predictive models currently in use rely on centralized datasets, necessitating the sharing of private patient information between institutions. This study uses the OPTN transplant dataset to propose a federated learning system for one-year kidney graft failure prediction. The system mimics a multi-hospital setting in which five transplant centers work together to train a Multi-Layer Perceptron model while maintaining patient data locally at each facility. FedAvg, FedProx, and SCAFFOLD are three federated optimization algorithms that are assessed and contrasted with a centralized baseline model. According to experimental results, federated methods like FedAvg and FedProx preserve data privacy while achieving performance comparable to centralized training. These results show how federated learning can be used for predictive modeling in kidney transplant outcome analysis while maintaining privacy.**

***Index Terms*—Kidney Transplantation, Graft Survival Prediction, Federated Learning, OPTN Dataset, Healthcare Machine Learning, Privacy-Preserving**

I. Introduction

When compared to long-term dialysis therapy, kidney transplantation is widely acknowledged as the most effective treatment for patients with end-stage renal disease (ESRD), offering much better survival outcomes and quality of life [12]. Graft failure is still a significant issue in transplant medicine despite improvements in surgical techniques, immunosuppressive medications, and organ distribution regulations. Therefore, forecasting transplant outcomes is crucial for enhancing patient care, directing clinical judgment, and maximizing the distribution of limited donor organs. One-year graft survival is regarded as a crucial measure of transplant success among various clinical outcomes [10]. Serious complications, such as a return to dialysis, retransplantation, or an increased risk of death, are frequently caused by early graft loss within the first year. As a result, precise one-year graft failure prediction has emerged as a key goal in transplant research and clinical practice. Clinicians can modify treatment plans, enhance post-transplant monitoring, and improve overall transplant outcomes by identifying high-risk patients early [13].

Understanding the variables linked to graft survival has been greatly aided by large national transplant registries. Comprehensive clinical data on donor characteristics, recipient demographics, and transplant procedures can be found in databases like the Organ Procurement and Transplantation Network (OPTN) [14], United Network for Organ Sharing (UNOS), and the Scientific Registry of Transplant Recipients (SRTR). These datasets offer useful tools for examining transplant results and figuring out graft survival predictors. A number of significant risk factors, such as donor age, donor organ quality, immunological compatibility, and early post-transplant renal function, have been identified by prior research utilizing these registries [15].

Traditionally, centralized statistical and machine learning methods have been used to predict graft survival, combining patient data from several institutions into a single repository for model training. Transplant outcomes have been extensively analyzed using regression-based survival models, Cox proportional hazard models, and more recently, machine learning algorithms [16]. In order to increase the predictive accuracy for graft survival and patient outcomes, machine learning techniques have recently been applied to transplant registry data [17]. Although these centralized methods have enhanced predictive performance and yielded valuable insights, they necessitate the gathering and exchange of private patient data between institutions. Large-scale collaborative modeling is difficult in real-world healthcare settings because such centralized data sharing is frequently constrained by institutional data governance policies, privacy laws, and ethical considerations. Federated learning (FL) has become a promising paradigm for collaborative machine learning in privacy-sensitive domains in order to get around these restrictions. Multiple institutions can collaboratively train predictive models without directly exchanging raw data thanks to federated learning. Rather, a central aggregation server receives only the model parameters from each participating institution, which uses its own dataset to train a local model. These parameters are then combined by the server to create a global model that integrates information from all participating institutions [18]. While maintaining patient privacy and adhering to data protection laws, this decentralized learning system enables cooperative model development.

Federated learning has become more popular in healthcare applications in recent years, such as survival modeling across distributed clinical datasets, medical imaging analysis, and disease prediction [19]. Federated learning is still not widely used to predict kidney transplant outcomes, especially for one-year graft survival. The majority of current research either uses

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centralized datasets or concentrates on other clinical outcomes like healthcare disparities or dialysis survival [6]. Inspired by these difficulties, this study uses the OPTN transplant dataset to examine the application of federated learning for one-year kidney graft failure prediction. The suggested framework makes use of actual multi-center data from the OPTN dataset, where the CTR_CODE variable is used to identify transplant centers. To build a federated learning environment for cooperative model training while maintaining patient privacy, the five centers with the most transplant records were chosen. A Multi-Layer Perceptron (MLP) model is trained using pre-transplant clinical features obtained from the OPTN registry by each participating hospital, which keeps its own local dataset. Only parameter updates are shared by the locally trained models with a central server. The parameters are then aggregated using federated aggregation algorithms and sent to the global model for updating. The study intends to preserve the advantages of multi-institutional data learning while enabling privacy-preserving collaborative modeling for kidney transplant outcome prediction.

A. Contribution

- **Federated Learning System for Graft Survival Prediction:** This study proposes a federated learning system to predict one-year kidney graft failure using the OPTN transplant dataset while ensuring that sensitive patient data remains within individual hospitals.
- **Simulation of a Multi-Hospital Learning Environment:** A realistic multi-institutional federated setup is constructed by treating five transplant centers as independent clients, enabling collaborative model training across distributed healthcare institutions.
- **Evaluation of Multiple Federated Optimization Algorithms:** The study implements and compares three widely used federated algorithms FedAvg, FedProx, and SCAFFOLD to analyze their effectiveness in handling heterogeneous clinical data.
- **Clinically Relevant Feature-Based Modeling:** The proposed model utilizes 14 pre-transplant clinical features derived from donor, recipient, and transplant characteristics to predict graft survival while avoiding label leakage.
- **Comparison with Centralized Learning Approaches:** A centralized baseline model is developed and compared with federated learning approaches to evaluate the trade-off between predictive performance and privacy-preserving collaborative learning.

B. Organisation

This is how the rest of the paper is structured. The relevant research on kidney transplant survival prediction and federated learning techniques in healthcare is presented in Section 2. The dataset preparation procedure, which includes data collection and preprocessing, is covered in Section 3. Four additional subsections are included in this section: (A) Data Source and Collection; (B) Feature Selection; (C) Outcome Definition; and (D) Federated Dataset Construction. Federated Learning Architecture and Federated Training Procedure are the two subsections that make up Section 4, which presents the suggested model and methodology. The experimental findings and discussion, which assess the effectiveness of various federated learning algorithms, are presented in Section 5. The paper is finally concluded in Section 6, which also suggests future research directions.

II. Related Work

Pruett et al. [1] used the US national transplant database (2000–2018) to investigate the relationship between donor age, KDPI, and graft survival. The study assessed factors that affect graft durability using survival analysis and one-year eGFR as a functional indicator. The findings demonstrated that post-transplant kidney function is more closely associated with graft survival than donor age.

In order to examine long-term survival and death with graft function (DWGF), Ojo et al. [2] examined registry data from 86,502 US recipients of renal transplants between 1988 and 1997. The study found variables linked to DWGF using population-based survival analysis. The findings demonstrated that cardiovascular disease is the primary cause of death and that DWGF contributes significantly to graft loss.

Hariharan et al. [3] examined the effect of first-year renal function on long-term graft survival by analyzing UNOS/OPTN data of 105,742 kidney transplant recipients (1988–1998). The study evaluated post-transplant renal performance using serum creatinine at six months and one year as predictors. The findings demonstrated that early renal function is a strong predictor of long-term graft survival and that elevated creatinine levels considerably raise the risk of graft failure.

In order to investigate the connection between one-year renal dysfunction and long-term graft failure, Siddiqi et al. [4] examined OPTN/UNOS data of 117,501 kidney transplants. The study assessed risk factors related to donors, recipients, and transplants using logistic regression and Cox proportional hazard models. The findings demonstrated that long-term graft failure is strongly predicted by elevated serum creatinine at one year.

In order to assess the effect of HLA matching and mismatching on kidney transplant outcomes, Yacoub et al. [5] examined OPTN/UNOS registry data (1995–2012). The study investigated their relationship with acute rejection, delayed graft function, and graft survival using multivariable statistical models. The findings demonstrated that compared to HLA mismatches, the number of HLA matches is a more reliable independent predictor of graft survival.

To assess patterns in long-term graft survival, Poggio et al. [6] examined SRTR data of 331,216 kidney transplants (1995–2017). The study evaluated changes in graft failure risk and survival outcomes using survival analysis across various transplant eras. Over time, the median graft survival improved and the risk of graft failure steadily decreased, according to the results.

Using transplant data from 1987 to 2010, Kachouie and Hamad [7] investigated how the 2003 OPTN policy affected kidney graft survival. The study examined the impact of donor characteristics and policy modifications on transplant outcomes using Kaplan–Meier and Cox proportional hazards models. After the OPTN policy was put into place, the results showed a decrease in graft failure rates and an increase in transplants.

Using registry data from 29,468 patients from 149 transplant centers, Tong et al. [8] proposed a federated learning framework to examine racial disparities in kidney graft failure. The study used counterfactual analysis and federated modeling to assess how the site of care affected graft outcomes. The findings demonstrated that variations in graft failure rates are influenced by variations in transplant center practices.

Using data from the NephroPlus dialysis network in India, Raju and Vepakomma [9] used federated learning to predict hemodialysis patients' survival outcomes. In order to facilitate cooperative learning across several centers without exchanging sensitive data, the study created federated survival prediction models. In comparison to locally trained models, the results demonstrated better predictive performance while maintaining data privacy.

To investigate trends in graft survival, Hariharan et al. [10] examined data from 93,934 kidney transplants performed in the US between 1988 and 1996. The study assessed the impact of transplant, post-transplant, and demographic factors on graft outcomes using regression-based survival analysis. Both short-term and long-term graft survival significantly improved over time, according to the results.

Using decentralized time-to-event data, Wang et al. [11] created a communication-efficient federated learning algorithm (CEDAR-t2e) to investigate racial differences in post-transplant survival. The approach was assessed using data from 73 transplant centers and 39,043 patients from the United States Renal Data System. The findings demonstrated the efficacy of federated learning for multi-center healthcare analysis while maintaining data privacy and revealed negligible site-of-care associated disparities.

A systematic review and meta-analysis were carried out by Ravindhran et al. [20] to assess the application of machine learning models for graft survival prediction in kidney transplantation. The study evaluated the diagnostic performance of different machine learning techniques in comparison to conventional regression methods by analyzing 31 research articles. The results indicated that machine learning models can improve graft survival prediction by effectively capturing complex nonlinear relationships in transplant data.

Using transplant data from 378 patients, Moghadam and Ahmadi [21] developed a machine learning framework to predict kidney graft rejection. The study introduced a clustering-based undersampling technique based on the Red Deer Algorithm (RDA) to address the issue of class imbalance. The effectiveness of the suggested method for graft rejection prediction was demonstrated by experimental results, which revealed that the decision tree classifier produced the best predictive performance.

Using data from 407 kidney transplant cases, Badrouchi et al. [22] examined the application of AI and machine learning techniques to predict long-term kidney transplant survival. The study identified important predictors of graft outcomes by contrasting several AI-based models with conventional statistical analysis. The results showed that machine learning models achieved high predictive performance, demonstrating the potential of AI-based approaches for supporting clinical decision-making in kidney transplantation.

Yi et al. [23] created a deep learning-based framework to predict graft loss by identifying pathological abnormalities in kidney transplant biopsy images. The study quantified tissue damage indicators and analyzed histopathological features from transplant biopsies using a deep learning pipeline. The findings showed that the suggested model outperformed conventional clinical scoring systems in terms of predictive accuracy, underscoring the potential of deep learning for risk assessment and graft loss prediction.

III. Data Collection and Preprocessing

A. Data Source and Collection

The Organ Procurement and Transplantation Network (OPTN) national transplant registry [14] provided the dataset used in this study, which includes extensive data on kidney transplantation practices nationwide. The raw dataset file of Kidney_transplant had a total file size of 1.2 GB and contained about 1.24 million records. The pandas library's chunk-based reading strategy (pd.read_csv with chunksize=100000) was used to load the dataset in order to facilitate large-scale data processing without memory overflow. Effective sequential loading and data aggregation were made possible by this method.

At first, the dataset included a variety of records, such as transplant events, waitlist entries, and status updates. Records were filtered to keep only transplant-related cases in order to concentrate on transplant procedures. To do this, observations having a legitimate transplant date (TX_DATE) were chosen. 580k+ kidney transplant cases were left for additional examination after this filtering stage.

A number of other filtering criteria were used in order to keep the study population uniform. Initially, cases of multi-organ transplants were eliminated because their clinical risk profiles were different from those of isolated kidney transplants. Second, because juvenile transplant outcomes include distinct clinical considerations, only adult recipients were kept. Only pertinent adult kidney transplant cases appropriate for predictive modeling were included in the dataset thanks to these preprocessing procedures.

B. Feature Selection

Recipient, clinical, donor, and transplant-related variables were among the 14 clinically significant pre-transplant characteristics chosen for modeling. Clinical indicators included dialysis status, panel reactive antibody (PRA) level, and transplant list waiting time; recipient characteristics included age,

gender, ethnicity, BMI, and diabetes status. Cold ischemia time and donor-recipient distance were transplant-related variables, while donor characteristics included donor age, donor type, KDPI, and donor creatinine. To guarantee that the model only uses pre-transplant data to predict graft survival, variables like post-transplant measurements and early rejection indicators that might cause label leakage were eliminated.

C. Outcome Definition

The primary prediction target in this study is 1-year graft failure, represented by the binary variable label_fail_1y. A transplant was labeled as failure if any of the following events occurred within 365 days after transplantation:

1) Graft Failure Date (FAILDATE_KI)
2) Retransplantation Date (RETXDATE_KI)
3) Return to Dialysis (RESUM_MAINT_DIAL_DT)
4) Patient Death (PX_STAT_DATE)

The earliest occurrence among these events was treated as the failure event date. If the event occurred within one year of the transplant date (TX_DATE), the label was assigned a value of 1, otherwise 0. Using this clinically meaningful definition, the dataset exhibited a 1-year graft failure rate of 15.9%(Approximately 98k+ samples), which provides a reasonably balanced distribution for machine learning modeling.

D. Federated Dataset Construction

To create a realistic multi-institutional federated learning environment, each transplant center was treated as a separate client. The dataset has a transplant center identifier (CTR_CODE) that matches different hospitals. The federated learning environment was built using the five centers that had the most transplant cases. In total, these centers provided 48,112 transplant records, which were sent to five separate clients that represented different hospitals. In the real world, healthcare systems are decentralized, so patient data can't be freely shared between hospitals. Each hospital has its own local dataset. This setup lets you use and test federated learning algorithms while keeping your data private.

The entire pipeline of the suggested federated learning system for kidney graft survival prediction using the OPTN transplant dataset is shown in Figure 1. The OPTN kidney dataset, which includes clinical, donor, and transplant-related characteristics, is where the process starts. In order to eliminate missing values, normalize variables, and create useful predictors, the dataset initially goes through data preprocessing, which includes cleaning, filtering, and feature engineering. Following preprocessing, the dataset is divided into several clients, each of which represents one of five hospitals. This simulates a decentralized healthcare setting in which each hospital keeps its local data. Without exchanging raw patient data, each client uses its own subset of data to separately train a local machine learning model.

A federated learning server receives the trained local model parameters and uses federated optimization algorithms like FedAvg, FedProx, and SCAFFOLD to aggregate them. These algorithms preserve data privacy by combining the locally

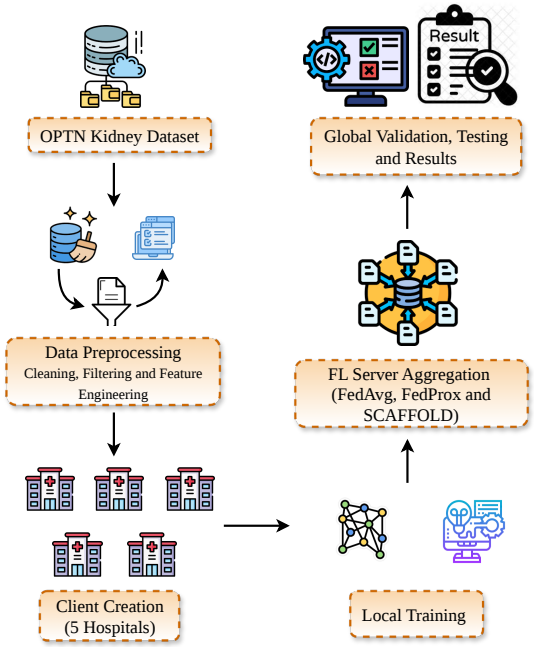


Fig. 1. Federated learning workflow for kidney graft survival prediction using the OPTN dataset.

acquired parameters and sending them to the global model for updating. Lastly, global validation and testing are used to assess the combined global model, and model performance metrics like ROC-AUC, PR-AUC, Macro-F1 score, and balanced accuracy are calculated.

IV. Proposed Work and Methodology

A. Federated Learning Architecture

The suggested federated learning architecture for privacy-preserving kidney graft survival prediction utilizing multi-institutional transplant data is depicted in Figure 2. Each of the five participating hospitals in the system maintains a local dataset generated from the OPTN transplant registry, simulating a decentralized healthcare setting. This design ensures compliance with healthcare data privacy regulations by preventing sensitive patient data from ever leaving the original institution.

Using its locally accessible transplant information, each institution individually trains a Multi-Layer Perceptron (MLP) model. To assess the likelihood of a 1-year graft failure, the MLP model learns predictive patterns from recipient characteristics, donor features, and transplant-related factors. This configuration represents a realistic non-IID federated learning scenario frequently seen in multi-center healthcare systems because the datasets across hospitals may have distinct distributions.

Every client uses its own dataset to execute local model updates during training. Only the model parameters or gradients are sent to a centralized server rather than the raw data.

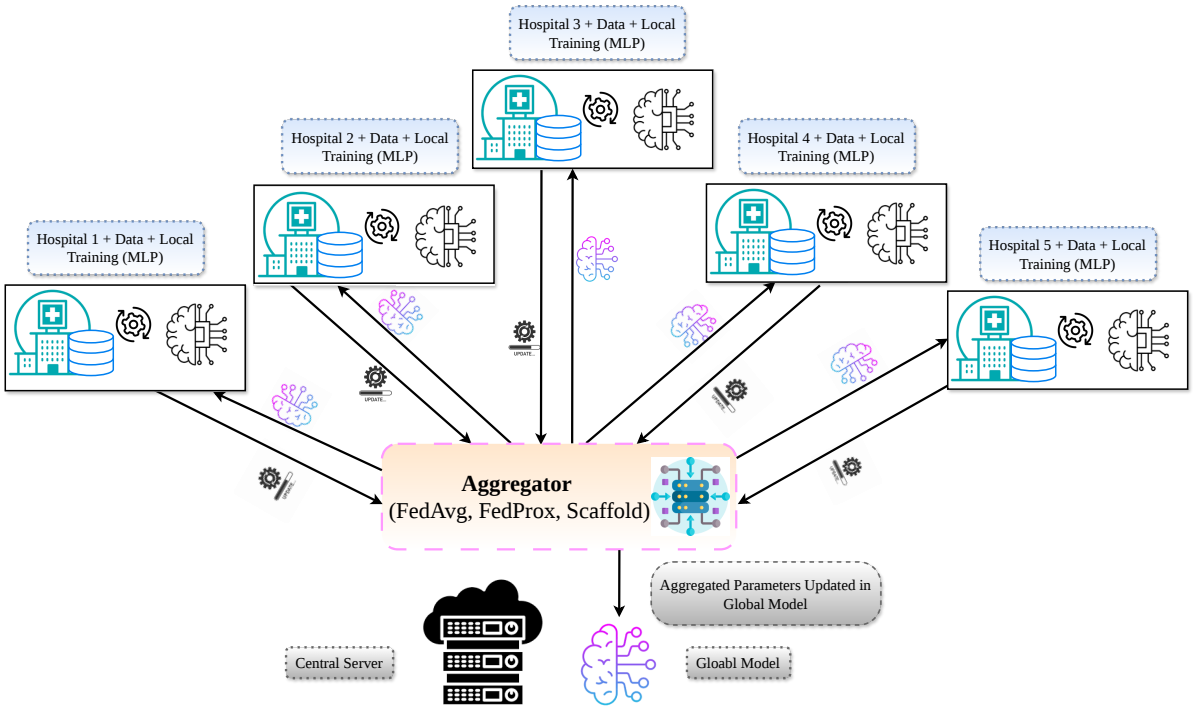


Fig. 2. Federated learning architecture illustrating decentralized training across multiple hospitals and centralized model aggregation.

While allowing for cooperative model creation across several healthcare organizations, this decentralized learning paradigm dramatically lowers privacy issues. A central aggregation server receives the updated parameters from all participating hospitals' local models, as shown in Figure 2, and aggregates them to create a global model. Federated optimization algorithms like FedAvg, FedProx, and SCAFFOLD are used in the aggregation process to help the server address data heterogeneity and communication efficiency while integrating knowledge gained from several clients.

B. Federated Training Procedure

There are several communication rounds in the federated training process. The global model parameters are first set by the central server, which then distributes the model to each participating institution. The Multi-Layer Perceptron (MLP) model then learns patterns unique to the local data distribution throughout a predetermined number of epochs of local model training conducted by each hospital using its proprietary dataset. The hospitals send their revised model parameters to the central server after finishing local training. Crucially, data privacy is ensured by sharing only model parameters and keeping the raw patient data inside each institution. The received parameters are then combined by the central server using federated optimization algorithms to carry out federated aggregation. Three popular techniques were assessed in this study: FedAvg (weighted averaging of client model parameters); FedProx (introducing a proximal regularization term to stabilize training under data heterogeneity); and SCAFFOLD (reducing client drift through control variates during optimization). After aggregation, all participating hospitals' knowledge is included into the global model by the server. The next round of local training is then started by redistributing the updated global model to each hospital. Until the model converges or the predetermined number of training rounds is attained, this iterative procedure is repeated for further communication rounds.

V. Result and Discussion

TABLE I
PERFORMANCE COMPARISON BETWEEN CENTRALIZED AND FEDERATED LEARNING MODELS FOR 1-YEAR GRAFT SURVIVAL PREDICTION.

Model	Training Type	ROC-AUC	PR-AUC	Macro-F1	Balanced Accuracy
Centralized MLP	Centralized	0.635	0.209	0.564	0.571
FedAvg	Federated	0.630	0.203	0.557	0.554
FedProx	Federated	0.630	0.201	0.555	0.553
SCAFFOLD	Federated	0.564	0.159	0.524	0.523

Table I compares the predictive performance of centralized and federated learning models for 1-year kidney graft survival prediction. The centralized Multi-Layer Perceptron (MLP) model performed the best overall, with ROC-AUC of 0.635, PR-AUC of 0.209, Macro-F1 score of 0.564, and balanced accuracy of 0.571. These findings imply that because all available data is directly accessible, training the model on a fully centralized dataset offers marginally better predictive capability.

Five participating clients (transplant centers) participated in the federated learning experiments, and each client completed 10 local training epochs over the course of 20 communication rounds. A batch size of 1024 and a learning rate of 1×10^{-3} were used to train the models.

FedAvg and FedProx performed similarly among the federated techniques, obtaining ROC-AUC values of 0.630, comparable Macro-F1 scores of 0.55, and balanced accuracy scores of 0.55. This suggests that federated learning can preserve data privacy across institutions while achieving performance comparable to centralized training. SCAFFOLD, on the other hand, performed comparatively poorly, with a ROC-AUC of 0.564 and balanced accuracy and macro-F1 of 0.523, indicating that it was less successful for this prediction task under the current experimental setup. Balanced accuracy was used instead of standard accuracy because the dataset exhibits class imbalance between graft survival and failure cases. Unlike normal accuracy, balanced accuracy accounts for both classes equally by averaging sensitivity and specificity, providing a more reliable evaluation of model performance.

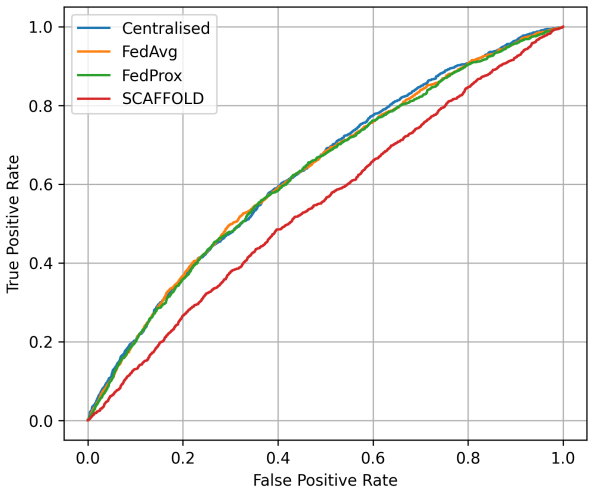


Fig. 3. ROC curve comparison of centralized and federated models

The classification performance of the centralized model and the federated learning techniques (FedAvg, FedProx, and SCAFFOLD) is compared in the ROC curve shown in fig. 3. The ability of the centralized model, FedAvg, and FedProx to differentiate between graft failure and non-failure instances is equivalent across the majority of criteria. The SCAFFOLD model, on the other hand, performs worse since its curve is constantly lower than that of the other approaches. Overall, the findings imply that federated strategies, such FedAvg and FedProx, can attain performance that is comparable to the centralized model.

Because of the class imbalance in graft failure prediction, the Precision–Recall (PR) curve displayed in fig. 4 offers a more informative evaluation for this investigation. FedAvg and FedProx are closely behind the centralized model, which retains the maximum precision over the majority of recall levels. The SCAFFOLD model performs poorly in accurately

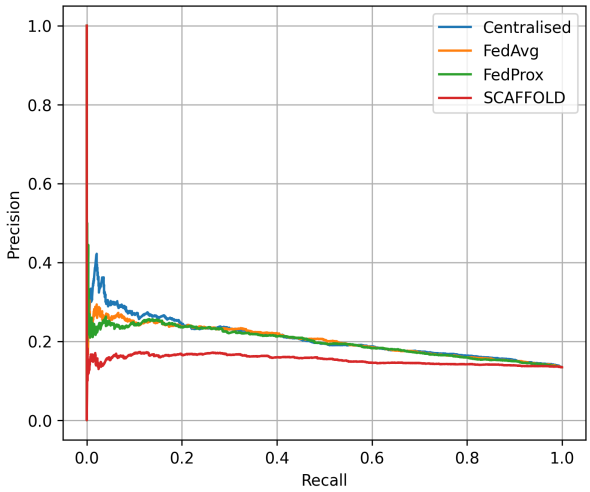


Fig. 4. Precision–Recall curve comparison of centralized and federated models

detecting positive cases, as evidenced by its reduced precision throughout the curve. These findings further demonstrate that FedAvg and FedProx maintain the advantages of federated learning while offering competitive performance when compared to centralized training.

VI. Conclusion

Using the OPTN transplant dataset, a federated learning system for predicting one-year kidney graft survival is presented in this work. The suggested method protects patient data privacy while facilitating cooperative model development across several institutions. According to experimental data, federated algorithms like FedAvg and FedProx outperform the centralized Multi-Layer Perceptron (MLP) model in a number of assessment measures, such as ROC-AUC, PR-AUC, Macro-F1 score, and balanced correctness. Conversely, the SCAFFOLD algorithm performed rather poorly with the given setup. Overall, the findings demonstrate federated learning's promise as a privacy-preserving approach for cooperative healthcare modeling, especially in settings where direct data exchange between institutions is prohibited. To further assess the scalability of the suggested federated learning system, future research will concentrate on adding more transplant centers and bigger multi-institutional datasets. To enhance predictive performance for kidney graft survival prediction, more sophisticated deep learning architectures and sophisticated federated optimization strategies will also be investigated.

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